

ME416

Ground Vehicle Dynamics

Kinematics,
Weight Transfer and ARB

Fall 2020

[1]

Lateral Load Transfer

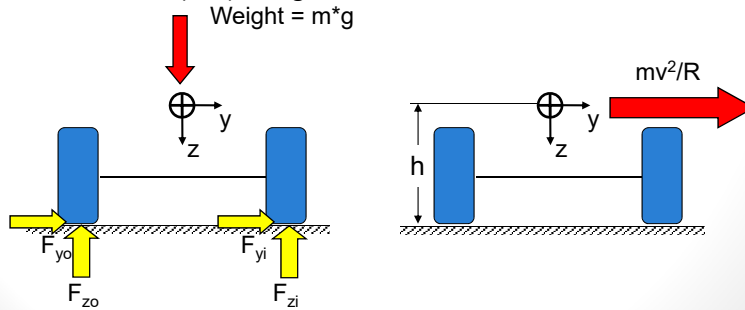
- As we have seen, the lateral force generated by a tire depends
 - The normal load on the tire due to
 - Static weight distribution
 - Dynamic weight distribution due to lateral and longitudinal forces
 - Tire properties due to tire material and construction
- We know that the distribution of forces on a mechanism depends on the kinematics of the mechanism, i.e. roll axis and c.g. location
- We also know that vehicle performance and stability is a function of lateral and longitudinal weight distribution – static and dynamic
- We will concentrate on lateral load transfer today

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Lateral Load Transfer FBD and MAD

- Assume a simple 2-D vehicle model
- Assume the vehicle is in equilibrium
 - No rotation about its roll axis – i.e. no angular acceleration
 - No vertical or longitudinal acceleration
- As shown the vehicle is negotiating a right hand turn (viewed from the rear)
- The roll center (axis) is at ground level

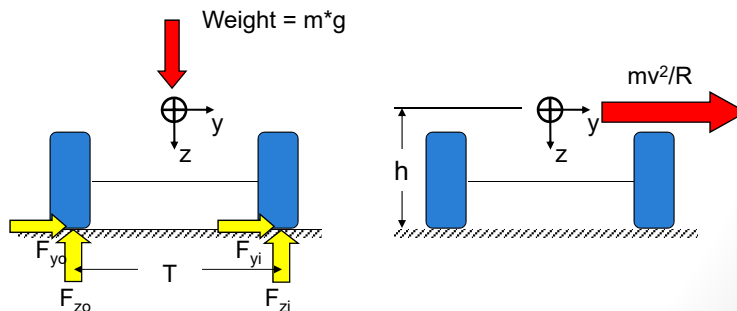


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[3]

Lateral Load Transfer FBD and MAD (2)

- The tire lateral forces are a function of the tire normal load and tire properties – i.e. nonlinear
- Choose moment center at the tire ground contact point to avoid the need to determine the tire lateral forces



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Load Transfer – Equilibrium Equations

- From the FBD and MAD, sum the moment about the inside tire

$$\sum M_i = -F_{zo}T + mg\frac{T}{2} = -hm\frac{v^2}{R}$$

- This yields the outside normal force as

$$F_{zo} = mg\left(\frac{1}{2} + \frac{v^2 h}{RgT}\right)$$

- Similarly we can sum the moment about the outside tire and obtain the inside tire normal force

$$\sum M_o = -F_{zi}T - mg\frac{T}{2} = -hm\frac{v^2}{R}$$

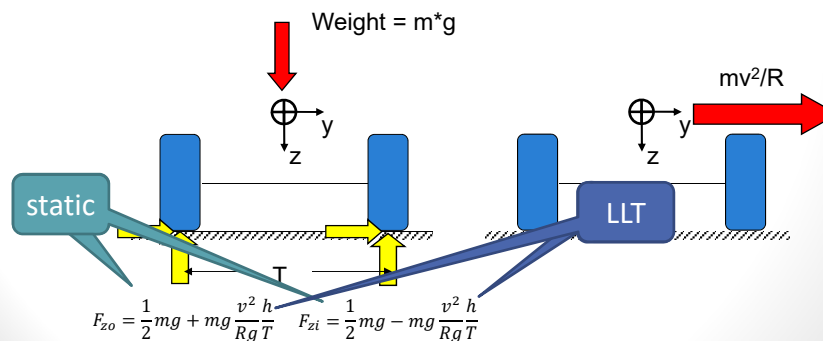
$$F_{zi} = mg\left(\frac{1}{2} - \frac{v^2 h}{RgT}\right)$$

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Stop Here and Think About What This Says

- The change in normal force is proportional to the vehicle's mass, height of the c.g. and lateral acceleration (v^2/R) and as expected the increase in normal load on the outside tire relative to static is equivalent to the reduction in normal load on the inside tire
- This is called the lateral load transfer (LLT) due to lateral acceleration



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Stop Here and Think About What This Says

- From these equilibrium equations we see that the total lateral load transfer is defined only by the vehicle mass, height of the c.g., track and the lateral acceleration
- What is the limit of lateral load transfer?
 - Overturn, i.e. when the inside wheel load(s) go to zero the system is unstable and overturn is likely
 - We will revisit overturn (rollover) in a later topic

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Lateral Weight Transfer - Components

- From the previous equilibrium equations we saw that the total lateral load transfer is defined only by the
 - Vehicle mass
 - C.G. height
 - Track width
 - Lateral acceleration
- However, how this lateral load transfer occurs or is transferred is determined by the geometry and mass configuration of the vehicle

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Lateral Load Transfer – Components (2)

- Recall,
 - Roll Center (RC): The point at which the lateral forces at the contact patch(es) may be applied to the sprung mass without producing an angular (roll) displacement, i.e. the force resolution point of the suspension on the sprung mass
 - Neutral Roll Axis: The line joining the front and rear RC's
 - Roll Rate or Stiffness: The change in the restoring couple exerted by the suspension at either pair of wheels on the sprung mass per unit change in roll angle of the sprung mass about a horizontal axis

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Lateral Load Transfer – Components (3)

- We are going to look at a simplified steady-state case to familiarize ourselves with the components of lateral load transfer
- We will make the following assumptions,
 - A horizontal lateral load applied along the neutral roll axis produces no roll of the sprung mass
 - Front and rear roll rates are measured independently
 - Tire deflection rates are included in the front and rear roll rates
 - Solid axle roll is neglected
 - Vehicle c.g. and RC are located on the vehicle centerline
- The vehicle is considered as three masses
 - Sprung mass
 - Chassis and body
 - Front and rear unsprung masses
 - Wheels, brakes, hubs, axles and some proportion of springs, control arms etc.

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Lateral Load Transfer – Components (4)

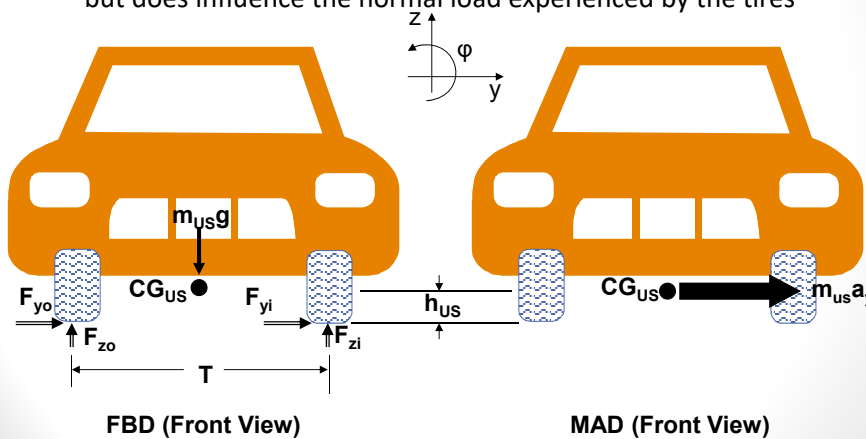
- If the vehicle is considered as three masses then we need to consider the load transfer of each of those three masses
 - Sprung mass
 - Chassis and body
 - Front and rear unsprung masses
 - Wheels, brakes, hubs, axles and some proportion of springs, control arms etc.

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Lateral Load Transfer – Unsprung

- The unsprung mass load transfer does not act on the body but does influence the normal load experienced by the tires



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Lateral Load Transfer – Unsprung Mass

- If we assume the unsprung mass to act at the axle height and sum the moments about the outside tire we have

$$\sum M_o = TF_{z_i} - \frac{T}{2}m_{US}g = -h_{US}m_{US}a_y$$

$$F_{z_i} = \frac{1}{2}m_{US}g - \frac{h_{US}}{T}m_{US}a_y$$

- Therefore the load transfer of the unsprung mass is

$$\Delta F_{z_{UWT}} = \pm \frac{h_{US}}{T}m_{US}a_y$$

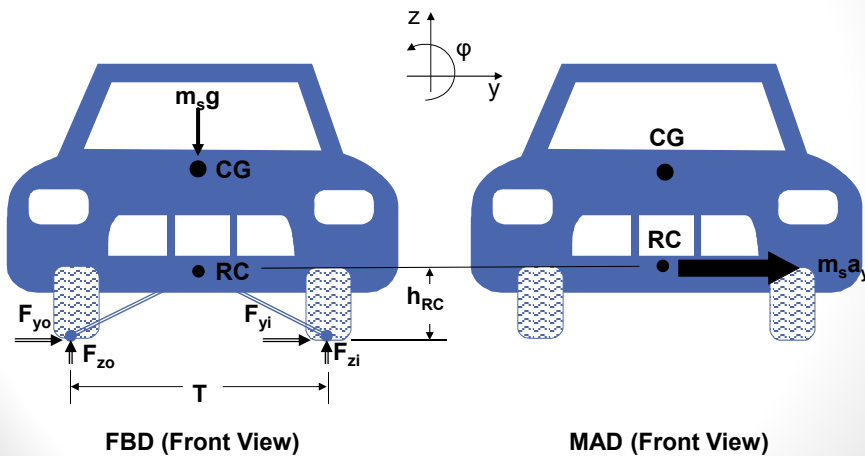
- Where the inside tire normal loading is reduced and the outside tire normal loading is increased
- Note that there will be an unsprung mass load transfer term for the front and rear axles
 - Or however many unsprung masses need to be accounted for

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Lateral Load Transfer – Geometric or Inelastic

- Also called the non-rolling overturning moment



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Lateral Load Transfer – Non-Rolling Overturning Moment

- A horizontal lateral load or acceleration applied along the neutral roll axis produces no roll of the sprung mass
- If we sum the moments about the outside tire we have

$$\sum M_o = TF_{z_i} - \frac{T}{2}m_s g = -h_{RC}m_s a_y$$

$$F_{z_i} = \frac{1}{2}m_s g - \frac{h_{RC}}{T}m_s a_y$$

- Therefore the geometric load transfer term is

$$\Delta F_{z_{GWT}} = \pm \frac{h_{RC}}{T}m_s a_y$$

- Where the inside tire normal loading is reduced and the outside tire normal loading is increased

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Lateral Load Transfer – Non-Rolling Overturning Moment (2)

- From the previous slide we developed the non-rolling overturning moment load transfer

$$\Delta F_{z_{GWT}} = \pm \frac{h_{RC}}{T}m_s a_y$$

- However, we have front and rear suspension with
 - Different roll centers
 - Different proportion of the sprung mass
- Therefore we need to modify the relationship to account for the front and rear suspension and mass distribution

$$F : \Delta F_{z_{GWT}} = \pm \frac{h_{FRC}}{T_F} \left(\frac{b}{WB} \right) m_s a_y$$

$$R : \Delta F_{z_{GWT}} = \pm \frac{h_{RRC}}{T_R} \left(\frac{a}{WB} \right) m_s a_y$$

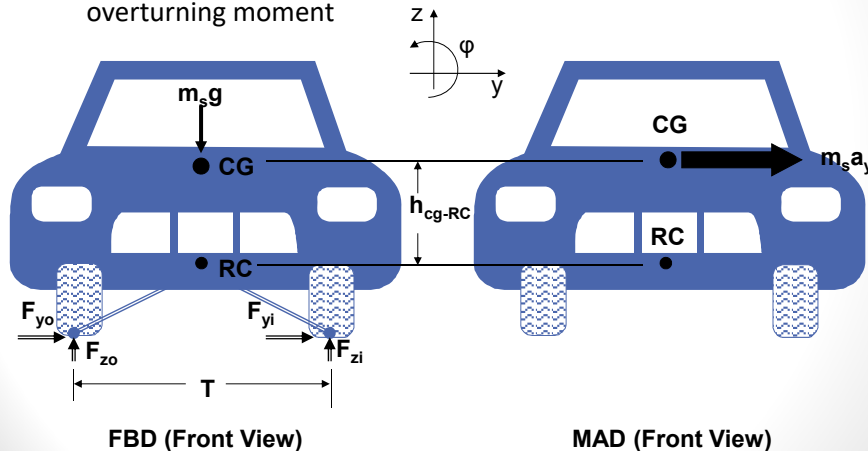
Where

- a = distance from the front axle to the c.g.
- b = distance from the c.g. to the rear axle line
- WB = wheelbase

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Lateral Load Transfer – Components: Elastic

- This is the only moment that results in body roll, i.e. the rolling overturning moment



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Lateral Load Transfer – Rolling Overturning Moment

- If we sum the moments about the outside tire we have

$$\sum M_o = TF_{z_i} - \frac{T}{2}m_s g = -h_{c.g.-RC}m_s a_y$$

$$F_{z_i} = \frac{1}{2}m_s g - \frac{h_{c.g.-RC}}{T}m_s a_y$$

- Therefore the elastic load transfer term is

$$\Delta F_{z_{EWT}} = \pm \frac{h_{c.g.-RC}}{T}m_s a_y$$

- Where the inside tire normal loading is reduced and the outside tire normal loading is increased

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Lateral Load Transfer – Rolling Overturning Moment (2)

- But we know that the total roll stiffness is the sum of the front and rear roll stiffnesses
 - What is the effect of the roll stiffness distribution?
- If we know the total roll stiffness of the suspension is

$$k_{\phi} = k_{\phi_f} + k_{\phi_r}$$

- The roll angle due to a given roll moment

$$\phi = \frac{h_{c.g.-RC} m a_y}{k_{\phi}}$$

- Where the moments on the front and rear are

$$M_{\phi_f} = k_{\phi_f} \phi$$

$$M_{\phi_r} = k_{\phi_r} \phi$$

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Lateral Load Transfer – Rolling Overturning Moment (3)

- The moments on the front and rear are
- $$M_{\phi_f} = k_{\phi_f} \phi$$
- $$M_{\phi_r} = k_{\phi_r} \phi$$
- These moments are applied through the suspension to the tires
 - Assuming the same track front and rear we can show the tire normal loads are changed from static by

$$\Delta F_{z_f} = \frac{k_{\phi_f} \phi}{T}$$

$$\Delta F_{z_r} = \frac{k_{\phi_r} \phi}{T}$$

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Lateral Load Transfer – Rolling Overturning Moment (4)

- These moments are applied through the suspension to the tires
 - Assuming the same track front and rear we can show the tire normal loads are changed from static by

$$\Delta F_{z_f} = \frac{k_{\phi_f} \phi}{T}$$

$$\Delta F_{z_r} = \frac{k_{\phi_r} \phi}{T}$$

- Noting that

$$\begin{aligned} \Delta F_{z_f} + \Delta F_{z_r} &= \frac{k_{\phi_f} \phi}{T} + \frac{k_{\phi_r} \phi}{T} \\ &= \frac{k_{\phi} \phi}{T} \\ &= \frac{h_{c.g.-RC} m a_y}{T} \end{aligned}$$

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Lateral Load Transfer – Rolling Overturning Moment (5)

- From the previous slide

$$\begin{aligned} \Delta F_{z_f} + \Delta F_{z_r} &= \frac{k_{\phi_f} \phi}{T} + \frac{k_{\phi_r} \phi}{T} \\ &= \frac{k_{\phi} \phi}{T} \\ &= \frac{h_{c.g.-RC} m a_y}{T} \end{aligned}$$

- What can we infer from this result?
 - Total load transfer is independent of the roll stiffness
 - However, the roll stiffness can be proportioned front to rear by adjusting the roll stiffness ratios
 - This allows us to proportion the lateral load transfer front/rear as well

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Lateral Load Transfer – Rolling Overturning Moment (6)

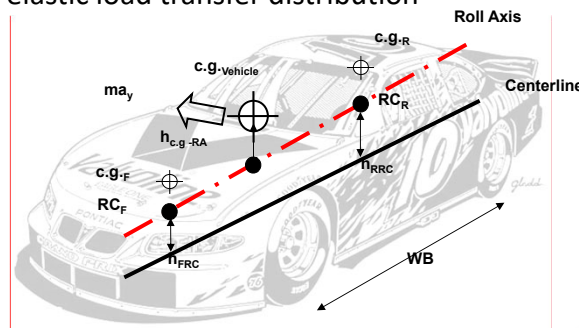
- From the previous slides we developed some simplified elastic load transfer relationships
- However, we generally have front and rear suspension with
 - Different roll centers
 - Different proportion of the sprung mass
- Also the rolling overturn moment acts on the sprung mass and
 - The roll stiffness for each axle is most likely not the same
- And as we have seen
 - The load transfer will be proportioned according to the roll stiffness proportioning

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Lateral Load Transfer – Rolling Overturning Moment (7)

- Recall that the roll centers define a roll axis
- We can define a height of the sprung mass c.g. above the roll axis to determine the overall rolling overturning moment
- We can use the roll stiffness distribution to determine the elastic load transfer distribution



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Lateral Load Transfer – Rolling Overturning Moment, General

- Without elaboration, the elastic load transfer terms are

$$F : \Delta F_{z_{EWT}} = \pm \frac{h_{c.g.-RA}}{T_F} \left(\frac{K_{\phi F}}{K_{\phi}} \right) m_s a_y$$

$$R : \Delta F_{z_{EWT}} = \pm \frac{h_{c.g.-RA}}{T_R} \left(\frac{K_{\phi R}}{K_{\phi}} \right) m_s a_y$$

- Where the total roll stiffness is given by

$$K_{\phi} = K_{\phi F} + K_{\phi R}$$

- The roll angle of the sprung mass is given by

$$\phi = m_s a_y h_{c.g.-RA} / K_{\phi}$$

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Lateral Load Transfer - Summary

- The elastic lateral load transfer is given by

$$F : \Delta F_{z_{EWT}} = \pm \frac{h_{c.g.-RA}}{T_F} \left(\frac{K_{\phi F}}{K_{\phi}} \right) m_s a_y$$

$$R : \Delta F_{z_{EWT}} = \pm \frac{h_{c.g.-RA}}{T_R} \left(\frac{K_{\phi R}}{K_{\phi}} \right) m_s a_y$$

- The geometric load transfer term is given by

$$F : \Delta F_{z_{GWT}} = \pm \frac{h_{FRC}}{T_F} \left(\frac{b}{WB} \right) m_s a_y$$

$$R : \Delta F_{z_{GWT}} = \pm \frac{h_{RRC}}{T_R} \left(\frac{a}{WB} \right) m_s a_y$$

- The unsprung load transfer is given by

$$\Delta F_{z_{UWT}} = \pm \frac{h_{US}}{T} m_{US} a_y$$

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Stop Here and Think About What This Says

- The elastic lateral load transfer is given by

$$F : \Delta F_{z_{EWT}} = \pm \frac{h_{c.g.-RA}}{T_F} \left(\frac{K_{\phi F}}{K_{\phi}} \right) m_s a_y$$

$$R : \Delta F_{z_{EWT}} = \pm \frac{h_{c.g.-RA}}{T_R} \left(\frac{K_{\phi R}}{K_{\phi}} \right) m_s a_y$$

- The elastic load transfer term shows that
 - As we change the roll stiffness for a particular axle we can adjust the elastic load transfer for that axle
 - However, we know that the total lateral load transfer is only a function of vehicle mass, lateral acceleration, c.g. height and track
- Therefore we cannot change the total lateral load transfer but we can adjust the fore/aft distribution of a portion of that load transfer by proportioning the front and rear roll stiffnesses

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Lateral and Longitudinal Weight Transfer

- So we know that vehicle performance and stability is a function of lateral and longitudinal weight distribution – static and dynamic
- We also just learned that the total lateral load transfer is fixed but we can adjust the fore/aft distribution of a portion of that load transfer by proportioning the front and rear roll stiffnesses
- Previously, we defined roll stiffness as a function of the effective spring rate and track only
 - However, we would like to isolate roll and heave stiffness
 - To allow us to independently tune ride and roll behavior.

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Lateral and Longitudinal Weight Transfer (2)

- We will define an anti-roll bar (ARB), or sway bar, as a torsion spring mounted transversely between the left and right wheels
 - Provides additional roll stiffness above the ride springs
- An ARB or equivalent mechanism can be used to proportion the lateral load transfer of the sprung mass
 - The ARB limits the body roll of the vehicle and also proportions the fore/aft roll stiffness distribution
 - This will also provide a tool for controlling the longitudinal, fore and aft, dynamic weight distribution
 - By adjusting the lateral and longitudinal load transfer we can control the normal load on each tire and tune the suspension to our equipment and/or needs

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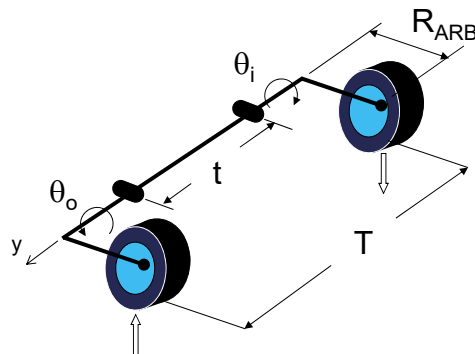
[29]

Anti-Roll Bar (ARB)

- The ARB is modeled as torsional spring with a resisting moment

$$M_{ARB} = k_{ARB}(\theta_o - \theta_i)$$

- We will assume that the ARB acts at the tire centerline



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Anti-Roll Bar (ARB) (2)

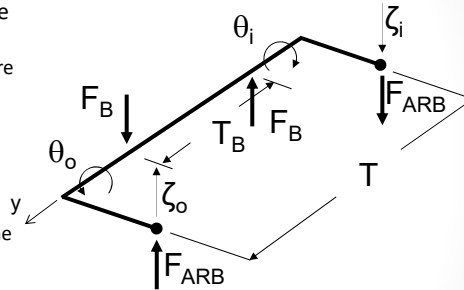
- The ARB is mounted to the sprung mass by bearings/bushes
- How is the moment transferred to the sprung mass and is it important?
 - The bearing/bushing forces (F_B) and tire forces (F_{ARB}) must form couples for equilibrium
- The moment applied to the chassis is

$$M_{\theta_{ARB}} = T_B F_B$$

- The equilibrium of the ARB in the roll plane requires

$$M_{\theta_{ARB}} = T_B F_B = -T F_{ARB}$$

- Therefore the ARB can be mounted conveniently on the chassis as the separation of the bearings or bushes have no effect on the moment transferred to the sprung mass

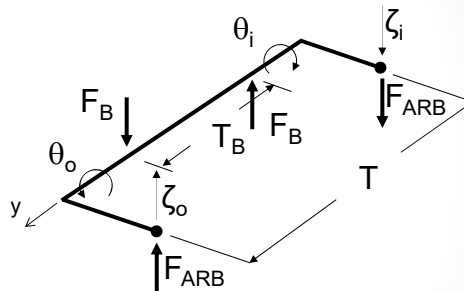


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Anti-Roll Bar (ARB) (3)

- Assumptions
 - Tire force assumed acts at tire center
 - ARB bearings assumed in tire plane for simplicity
 - ARB and bearing force of the same magnitude and treated as a spring force



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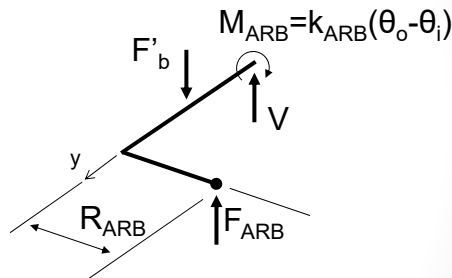
Vehicle Attitude and ARB

- We can see the ARB acts positively on the outboard side and negatively on the inboard side where

$$F_{ki} = k_r \left(\zeta - \frac{T}{2} \phi \right) - F_{ARB}$$

$$F_{ko} = k_r \left(\zeta + \frac{T}{2} \phi \right) + F_{ARB}$$

spring force due to bump and roll as developed in previous lecture



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ARB and Roll Angle

- Solving the spring force relationships with ARB terms for roll angle, ϕ , yields

$$F_{ki} = k_r \left(\zeta - \frac{T}{2} \phi \right) - F_{ARB}$$

$$F_{ko} = k_r \left(\zeta + \frac{T}{2} \phi \right) + F_{ARB}$$

$$\phi = \frac{1}{Tk_r} (F_{ko} - F_{ki} - 2F_{ARB})$$

- For the roll equation
 - ARB term is negative so the net effect on the sprung mass is to reduce the roll angle
- However, the tire normal load is
 - Reduced on the inboard tire
 - Increased on the outboard tire

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Lateral Weight Transfer – Rolling Overturning Moment Redux

- Our ARB finding on the previous slide is consistent with the relationship(s) developed earlier

$$\begin{aligned}\Delta F_{z_f} + \Delta F_{z_r} &= \frac{k_{\phi_f} \phi}{T} + \frac{k_{\phi_r} \phi}{T} \\ &= \frac{k_{\phi} \phi}{T} \\ &= \frac{h_{c.g.-RC} m a_y}{T}\end{aligned}$$

- What can we infer from this result?
 - Total lateral load transfer is independent of the roll stiffness
 - However, the roll stiffness can be proportioned front to rear by adjusting the roll stiffness ratios
 - This allows us to proportion the lateral weight transfer front/rear as well
 - For constant total roll stiffness
 - Increasing an axle's roll stiffness increases LLT
 - Reducing an axle's roll stiffness reduces LLT

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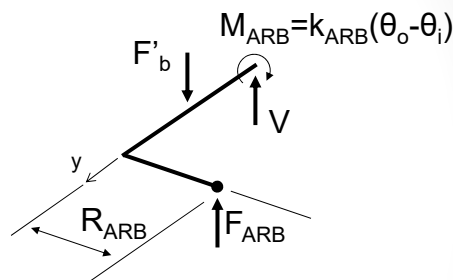
[35]

ARB and Roll Angle, Moment and Stiffness

- Sum moments about the ARB axis

$$R_{ARB} F_{ARB} - M_{ARB} = 0$$

$$\begin{aligned}F_{ARB} &= \frac{M_{ARB}}{R_{ARB}} \\ &= k_{ARB} \frac{(\phi_o - \phi_i)}{R_{ARB}}\end{aligned}$$



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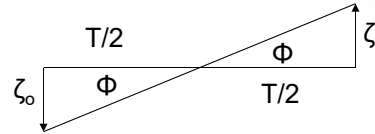
[36]

ARB and Roll Angle, Moment and Stiffness (2)

- We can relate the ARB twist and chassis roll where

$$\zeta = \zeta_o = -\zeta_i = \frac{T\phi}{2}$$

$$\phi = \phi_o = -\phi_i = \frac{\zeta}{R_{ARB}}$$



- Yielding

$$\phi = \frac{T\phi}{2R_{ARB}}$$

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ARB and Roll Angle, Moment and Stiffness (3)

- Combining, we find the ARB force contribution

$$F_{ARB} = \frac{k_{ARB}}{R_{ARB}} \left[\frac{T\phi}{2R_{ARB}} - \left(-\frac{T\phi}{2R_{ARB}} \right) \right] = \frac{k_{ARB} T\phi}{R_{ARB}^2}$$

- Roll moment contribution

$$M_{ARB} = TF_{ARB} = \frac{k_{ARB} T^2}{R_{ARB}^2} \phi$$

- Roll stiffness contribution

$$k_{\phi_{ARB}} = \frac{M_{ARB}}{\phi} = \left(\frac{T}{R_{ARB}} \right)^2 k_{ARB}$$

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Total Roll Stiffness

- By employing ARB's we can set the front roll stiffness (k_{ϕ_f}) and rear roll stiffness (k_{ϕ_r}) independently of the ride springs
- The total roll stiffness of the suspension is

$$k_{\phi} = k_{\phi_f} + k_{\phi_r} = (k_{\phi_{ride}} + k_{\phi_{ARB}})_f + (k_{\phi_{ride}} + k_{\phi_{ARB}})_r$$

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Addendum:

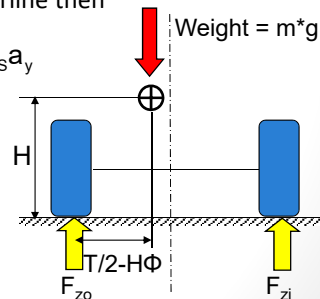
Weight Transfer – c.g. Lateral Displacement

- As the body rolls the c.g. is displaced laterally and will therefore have a contribution to the rolling moment
- The c.g. shift is difficult to determine without detailed suspension analysis
 - We can assume the displacement is small and if we assume that the vehicle rolls about the ground plane centerline then

$$\sum M_o = TF_{z_i} - \left(\frac{T}{2} - H\phi \right) W_{US} = -Hm_{US}a_y$$

- Solving for F_{z_i}

$$F_{z_i} = \left(\frac{1}{2} - \frac{H\phi}{T} \right) W_{US} - \frac{H}{T} m_{US} a_y$$



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Load Transfer – c.g. Lateral Displacement

- The weight transfer is the difference between the applied and static load

$$\begin{aligned}\Delta W &= F_{z_i} - \frac{1}{2} W_{US} \\ &= \pm \left(\frac{H\phi}{T} W_{US} + \frac{Hm_{US}a_y}{T} \right)\end{aligned}$$

- So the contribution to the weight transfer due to the c.g. shift is

$$\Delta F_{\zeta_{cg}} = \pm \frac{H\phi}{T} W_{US}$$